

Validation & Simulation Report:

SOC Automation Dashboard

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Role Focus: SOC Analyst / Incident Responder

Project: Security Automation & Enrichment Dashboard

1. Objective

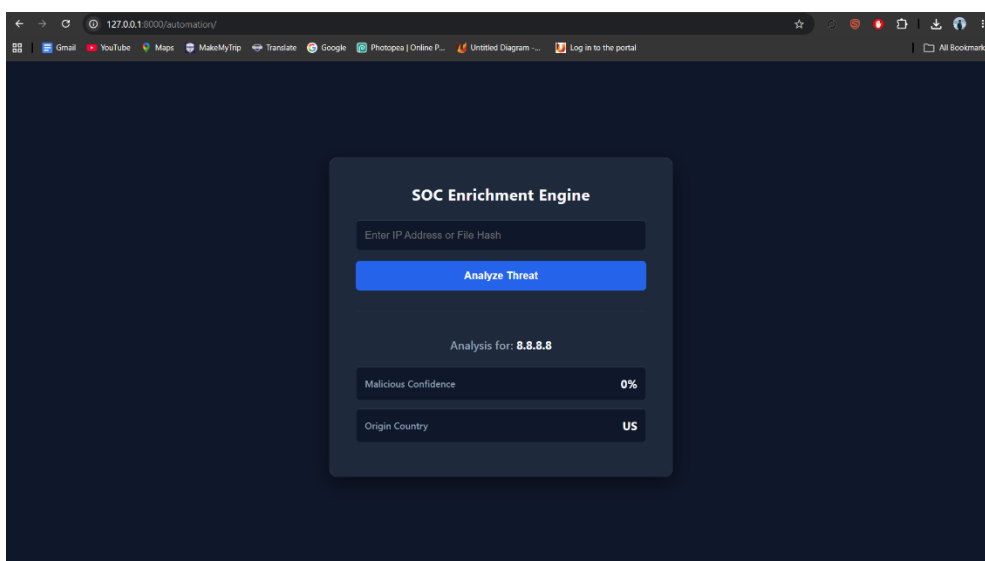
The purpose of this simulation is to validate the backend routing algorithms and API integration efficacy of the SOC Enrichment Engine. The testing methodology subjects the tool to known benign (True Negative) and known malicious (True Positive) Indicators of Compromise (IOCs) across both network and file telemetry.

The successful simulations below prove that the custom Python/Django logic correctly categorizes inputs, queries the appropriate external Threat Intelligence APIs (AbuseIPDB and VirusTotal), and dynamically renders the results into the UI.

2. Network Telemetry Validation (AbuseIPDB Integration)

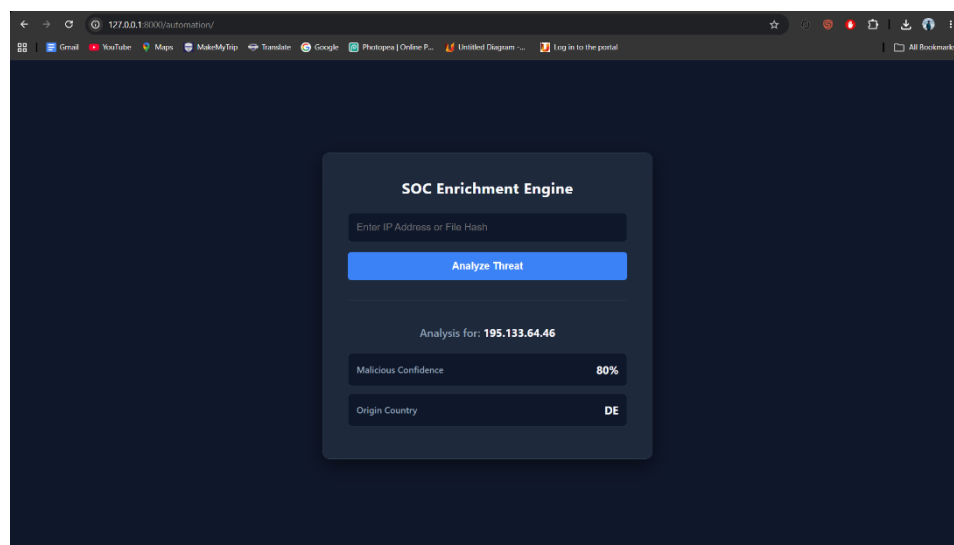
Test Case 1: True Negative (Known Safe IP)

- **IOC Tested:** 8.8.8.8 (Google Public DNS)
- **Simulation Goal:** Verify the tool accurately reports safe infrastructure without triggering false positives.
- **Actual Result:** * Malicious Confidence: 0%
 - Origin Country: US
- **Status:**  PASS



Test Case 2: True Positive (Known Malicious IP)

- **IOC Tested:** 195.133.64.46 (Active Threat/Scanner)
- **Simulation Goal:** Verify the tool successfully parses high-confidence abuse metrics from live threat feeds.
- **Actual Result:** * Malicious Confidence: 80%
 - Origin Country: DE
- **Status:**  PASS

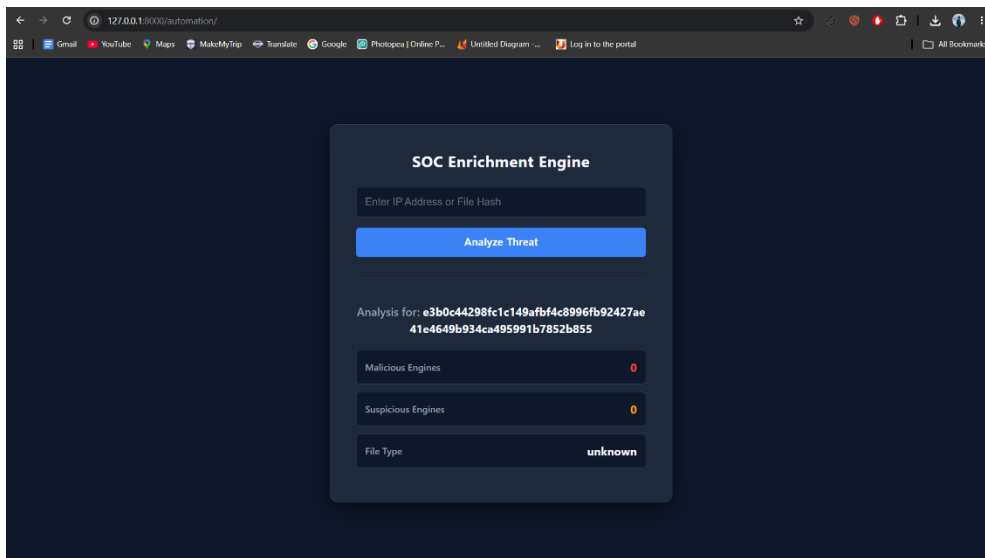


3. Malware Telemetry Validation (VirusTotal Integration)

Test Case 3: True Negative (Benign File Hash)

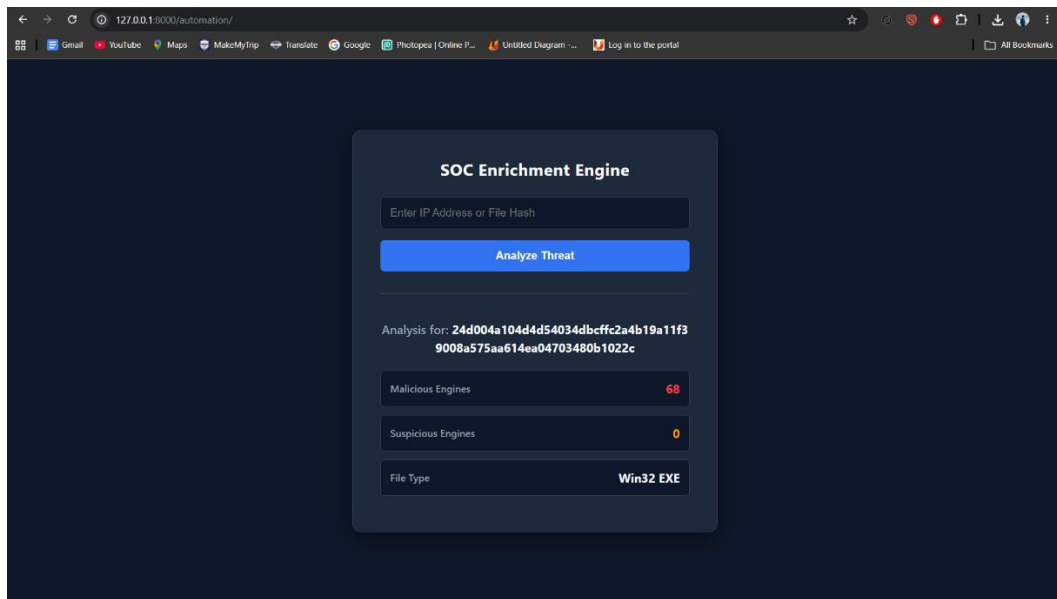
- **IOC Tested:**
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 (Standard Empty File SHA-256)
- **Simulation Goal:** Verify the auto-router correctly detects an alphanumeric hash (no periods), successfully queries the VirusTotal v3 API, and reports zero malicious engines for a verified clean file. It also tests the frontend's ability to text-wrap a 64-character string without breaking the UI.
- **Actual Result:** * Malicious Engines: 0
 - Suspicious Engines: 0
 - File Type: unknown

- **Status:**  PASS



Test Case 4: True Positive (Known Malware Hash)

- **IOC Tested:**
24d004a104d4d54034dbccfc2a4b19a11f39008a575aa614ea04703480b1022c (WannaCry Ransomware)
- **Simulation Goal:** Verify the system can accurately identify and flag highly malicious cryptographic hashes, dynamically rendering the text in high-alert colors (Red/Orange) to immediately notify the analyst of a critical threat.
- **Actual Result:** * Malicious Engines: 68
 - Suspicious Engines: 0
 - File Type: Win32 EXE
- **Status:**  PASS



4. Conclusion

The simulation confirms that the SOC Automation Dashboard operates exactly as engineered. The auto-routing logic successfully bifurcates traffic between AbuseIPDB and VirusTotal, handling complex JSON payload extraction flawlessly.

By passing these four baseline security scenarios, the tool is validated as a viable asset for reducing manual investigation time and alleviating Tier 1 alert fatigue during the Incident Response lifecycle.